Deep neural networks (DNNs)

Let us now look at how deep neural networks work.

The word 'neural' in DNN is related to biology. Actually, the soul of these networks is how the biological neural system works. So, let's take a brief look at how two biological neurons communicate.

There are three main parts in a biological neuron as shown in Figure 1.

1. Dendrite: This acts as input to the neuron from another neuron.
2. Axon: This passes information from one neuron to another.
3. Synaptic connection: This acts as a connection between two neurons. If the strength of the received signal is higher than some threshold, it activates another neuron.

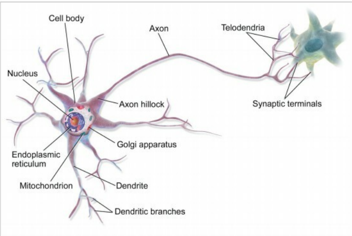

Figure 1: Biological neural network *(Source: https://en.wikipedia.org/wiki/Biological_neural_network)*

Neuron types

Let us try to express human decisions and biological neural networks mathematically so that computers can comprehend them.

Let's suppose that you want to go from city A to city B. Prior to making the journey, there are three factors that will influence your travel decision. These are:

a. If the weather (x1) is good (represented by 1) or bad (represented by 0) has a weight of (w1)
b. If your leave (x2) is approved (represented by 1), or not (represented by 0) has a weight of (w2)
c. If transport (x3) is available (represented by 1) or not (represented by 0) has a weight of (w3)

And you will decide as follows: Irrespective of whether your leave is approved or not and transport is available or not, you will go if the weather is good. This problem statement can be drawn as shown in Figure 2.

According to the figure, if the sum of the product of the inputs (xi) and their respective weights (wi) is greater than some threshold (T), then you will go (1), else you will not (0).

$$f(y) = \begin{cases} 0, & \text{where, } y = \sum_{i=1}^{3} X_i W_i, \text{ which is } \leq T \\ 1, & \text{where, } y = \sum_{i=1}^{3} X_i W_i, \text{ which is } > T \end{cases} \quad \textit{Where, f(y) is the decision} \dashrightarrow \text{eq(1)}$$

$$y = \sum_{i=1}^{3} X_i W_i = x_1 w_1 + x_2 w_2 + x_3 w_3 = 6 \cdot 1 + 2 \cdot 0 + 2 \cdot 0 = 6 > \text{threshold}(5) = \text{true} \quad \dashrightarrow \text{eq(2)}$$

$$w1 = w1 - \frac{\partial p}{\partial w1} \quad \dashrightarrow \text{eq(3)}$$

As your input and output is fixed, you have to choose weights and thresholds to satisfy the equation.

For example, let us choose w1=6, w2=2, w3=2 and T=5.

You will be able to make a correct decision if we choose the above values for equation (1), i.e., if your leave is not approved (0) and transport is not available (0) but the weather is good (1), then you should be going.

$$y = \sum_{i=1}^{3} X_i W_i = x_1 w_1 + x_2 w_2 + x_3 w_3 = 6 \cdot 1 + 2 \cdot 0 + 2 \cdot 0 = 6 > \text{threshold}(5) = \text{true}$$

Similarly, you can check other conditions as well.

It will be easy to manually calculate these weights and thresholds for small decision-making problems but as complexity increases, we need to find other ways – and this is where mathematics and algorithms come in.

f(y), the function represented in Figure 3, produces output in terms of 0 and 1. This says that for a small change in input, the change in output is high — represented by a step function. This can cause problems in many cases. Hence, we need to define a function f(y) such that for small changes in input, changes in the output are small — represented by the Sigmoid function.

Depending upon these conditions, there are two neuron types defined, as shown in Figure 3.

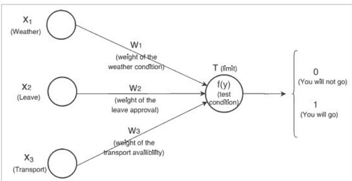

Figure 2: Simple human decision in mathematical form

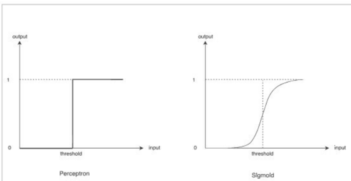

Figure 3: Neuron types